Esko Ilmari Vahteristo
[redacted] | i.vahteristo@gmail.com | **GitHub** | **LinkedIn** | **ilmarivahteristo.com**

SUMMARY

I am a product-minded ML/CV engineer (MSc) with ambition and curiosity. I am driven by hard challenges, learning, and impact. I've worked on a startup developing measurement systems for industry and manufacturing, and previously as a researcher at LUT University on ill-posed inverse problems. Along my work achievements, I've published academic papers, won competitions, and developed open-source software. I stay up-to-date on technologies and AI tools, and I am proficient in Python, Linux, Git, C, and the likes. I have a good foundation and I'm a fast learner.

WORK EXPERIENCE

Software engineer (ML/CV), Virelabs, 05/2025 - 8/2026

- Developed software for edge devices in industrial environments utilizing cameras, LiDARs, lasers. I also communicated with customers regularly, and worked closely with the hardware team.
- Notable tasks:
    1) 3D reconstruction of a moving vehicle using line cameras and LiDAR + radar. Paper.
    2) Quality control for veneer using laser triangulation. 0.5mm accuracy from >1m away.
    3) Object tracking with multiple cameras through occlusions, 5cm accuracy from 6m away.
    4) OCR system on industrial packaging with majority voting reaching 99.9% accuracy.

Research assistant, LUT University, 04/2024 – 08/2024

- First authored an IEEE conference paper on limited angle tomography using deep learning and image priors. paper

Freelance, Tmi Ilmari Vahteristo, 01/2024 - 04/2025

- Worked on reducing food waste at a restaurant by predicting customer volume based on menu.

OTHER TECH PROJECTS AND COMPETITIONS

Junction 2025, 11/2025

- Won Elevenlabs challenge for creating the best voice-based AI software. Demo

Research paper on numerical optimization for spherical codes, 02/2026 - 10/2026

- Developed a solver for spherical codes, beating the all-time record on 15% of cases. paper

LLM 20 questions, Kaggle, 06/2024 – 08/2024

- Ranked *7th out of 800* in creating an LLM based solution to play the "20 Questions" game.

Koodausta ja kisailua, Visma, 01/2023

- Won first place out of 20 competing teams. The task was to fix as many bugs as possible.

Google Foobar challenge, 05/2022 – 05/2023

- Nine difficult algorithmic problems. Got an interview at Google after completing.

IndySCC, CSC, 01/2022 – 11/2022

- Supercomputing competition, where we ranked 3rd from 10 universities across the world.

Reinforcement learning for a multiplayer card game, 10/2022 – 10/2023

- Trained an AI to play a card game with pure self-play, better than top humans. website, paper

EXTRA CURRICULAR

Project Manager, Lateksii, 01/2021 – 12/2022

- Organized events with our project team (4) + volunteers. Largest event 900 people.

Peacekeeper, UNIFIL, 11/2020 – 07/2021

- Squad co-lead, promoted part time to squad lead, and then team co-lead.

EDUCATION

- Graduated MSc (Tech), with 4.2 GPA, from Computer vision and pattern recognition, LUT university (08/2025) + exchange in Applied Math in Portugal, Lisbon, IST.
- 2nd MSc in Robotics and Automation engineering, Aalto University (estimated 9/2029)